

Cassels–Swinerton-Dyer conjecture for cubic surfaces

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Abstract

We resolve the Cassels–Swinerton-Dyer conjecture in dimension two: a smooth cubic surface over an arbitrary field with a zero-cycle of degree one has a rational point. The essential step is a descent from a linear Pfaffian presentation to a rational point in characteristic zero. A degree-two incidence of common isotropic three-spaces, together with a radical vector for a fourth alternating form, produces a point over an extension of degree at most two. Quadratic secant descent and specialization then give a point over the ground field. The classical five-point Pfaffian construction and Voisin’s degree reduction complete the characteristic-zero argument. Following Ma’s lifting method, finite subschemes and mixed-characteristic specialization extend the result to arbitrary fields.

Contents

1	Introduction	1
2	Small-degree points and twisted cubics	2
3	Pfaffian cubics have rational points	4
4	Five points and the characteristic-zero theorem	5
5	Specialization to arbitrary fields	6

1 Introduction

The Cassels–Swinerton-Dyer conjecture predicts that a cubic hypersurface which acquires a rational point over a finite extension of degree prime to three already has a rational point over its ground field. Equivalently, a zero-cycle of degree one should imply a rational point. This asks whether the numerical condition that the index be one is sufficient for the existence of a point. We prove the conjecture for smooth cubic surfaces.

Theorem 1.1. *Let k be an arbitrary field and let $S \subset \mathbf{P}_k^3$ be a smooth cubic surface. If S has a zero-cycle of degree one, then $S(k) \neq \emptyset$.*

Here zero-cycles have integral coefficients and need not be effective. Put $h = c_1(\mathcal{O}_S(1))$ and $h_3 = h^2 \in \mathrm{CH}_0(S)$, so that $\deg h_3 = 3$. A zero-cycle of degree one has in its support a closed point of degree prime to three. Conversely, the class of such a point, together with h_3 , gives a zero-cycle of degree one by Bézout’s identity. Thus Theorem 1.1 is equivalent to descent of rational points along finite extensions of degree prime to three.

The principal earlier reduction is due to Daniel Coray. Over a perfect field, a smooth cubic surface with a closed point of degree prime to three has a closed point of degree 1, 4, or 10 [5, Theorem 7.1]. Coray also proved the conjecture over local fields [5]. Qixiao Ma removed the perfectness hypothesis from the degree reduction, proving the same 1, 4, 10 conclusion for regular cubic surfaces over arbitrary fields [8, Theorem 2.6]. His argument lifts closed points as finite subschemes over a Cohen ring, thereby retaining their degrees even when the residue extensions are inseparable. Jean-Louis Colliot-Thélène developed the relation between such degree reductions and effectivity in the Chow group of zero-cycles on del Pezzo surfaces [4].

Claire Voisin recently sharpened Coray’s reduction in characteristic zero: a smooth cubic surface with a zero-cycle of degree one has a closed point of degree 1 or 4 [11, Theorem 1.5]. Her proof uses rank-two vector bundles and effectivity results for zero-cycles. The relation between degrees four and five is already present in her proof of Proposition 4.3(b). In a related direction, Francesca Balestrieri proved that a cubic hypersurface with a point over a simple extension of degree four has a point over an extension of degree one or five [1, Theorem 3.8]. For cubic surfaces in characteristic zero, Proposition 2.3 below gives both directions by intersecting the surface with a split twisted cubic.

Our additional arithmetic input is the following statement.

Theorem 1.2. *Let k have characteristic zero. If a nonzero cubic form $f \in k[x_0, x_1, x_2, x_3]$ has a linear 6×6 Pfaffian presentation over k , then the cubic $f = 0$ has a k -rational point.*

The geometry underlying this statement is classical. The construction of a Pfaffian presentation from five points in general linear position uses the structure theorem of David Buchsbaum and David Eisenbud [3]; it belongs to the theory of determinantal and Pfaffian hypersurfaces developed by Arnaud Beauville [2, Proposition 7.2 and Remark 7.3(b)] and is made explicit for cubic surfaces by Fabio Tanturri [10, Section 2]. The incidence of common isotropic three-spaces for three alternating forms on a six-dimensional space is generically of degree two, as proved by Frédéric Han [7, Proposition 2.4]. We give a short calculation of this incidence over the ground field. The arithmetic step is to choose a radical vector for the fourth form on one of these isotropic spaces. Four covectors then lie in a three-dimensional annihilator, producing a point of the Pfaffian cubic over an extension of degree at most two. Secant descent gives a rational point, and a one-parameter deformation removes the genericity condition.

Consequently, an effective zero-cycle of degree five on a smooth cubic surface in characteristic zero gives a rational point. The twisted-cubic construction and Voisin’s theorem establish Theorem 1.1 in characteristic zero. The passage to arbitrary characteristic follows Ma’s method: lift a closed point of degree prime to three as a finite flat subscheme, apply the characteristic-zero result, and specialize the resulting rational point by properness. The Hilbert-scheme input goes back to John Fogarty [6]; only formal smoothness at finite local complete intersections is needed for this last step.

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2 Small-degree points and twisted cubics

Throughout Sections 2–4, the field k has characteristic zero. The degree of a finite k -scheme is its length over k . We use the associated effective zero-cycle when taking residual intersections.

Lemma 2.1 (Quadratic secant descent). *Let $X \subset \mathbf{P}_k^3$ be the zero locus of a nonzero cubic form. If X has an effective zero-cycle of degree one or two, then $X(k) \neq \emptyset$.*

Proof. It suffices to consider a closed point P of degree two. Its two conjugates span a k -line ℓ . If $\ell \subset X$, it supplies a rational point. Otherwise $[\ell \cap X] - [P]$ is an effective zero-cycle of degree one. $\square$

Lemma 2.2. *Let P be a closed point of degree $d \in \{4, 5\}$ on a smooth cubic surface S/k .*

1. *If P is contained in a k -plane, then $S(k) \neq \emptyset$.*
2. *If P is not contained in a k -plane, its geometric points are in general linear position: any four of them span $\mathbf{P}^3$.*

Proof. For (1), let Π contain P . Since $h^0(\Pi, \mathcal{O}_\Pi(2)) = 6 > d$, there is a conic $Q \subset \Pi$ containing P . Write $C = S \cap \Pi$. If C and Q have no common component, then $[C \cap Q] - [P]$ is effective of degree $6 - d \leq 2$, and Lemma 2.1 applies. Otherwise the greatest common divisor of their equations over k has degree one or two. In the first case it defines a k -line on S ; in the second, the residual factor of the cubic defines such a line.

For (2), the assertion is immediate when $d = 4$. Suppose $d = 5$ and choose a finite Galois splitting field L/k . Normalize representatives $v_1, \dots, v_5 \in L^4$ by a common affine chart defined over k . Their span is L^4 , so their space of relations is one-dimensional. In a nonzero relation $\sum c_i v_i = 0$, the set of indices with $c_i = 0$ is Galois invariant. Transitivity forces this set to be empty, proving the assertion. The geometric span descends to k , so a proper span would indeed be contained in a k -plane. $\square$

Proposition 2.3. *A smooth cubic surface S/k has an effective zero-cycle of degree four if and only if it has an effective zero-cycle of degree five. More precisely, a closed point of either degree yields either a rational point or a closed point of the other degree.*

Proof. If $S(k) \neq \emptyset$, multiples of a rational point give both cycles. Otherwise Lemma 2.1 excludes points of degrees one and two; hence an effective zero-cycle of degree $d \in \{4, 5\}$ must be a single closed point P . By Lemma 2.2, its geometric points are in general linear position.

We place P on a *split* twisted cubic over k . Write $E = \kappa(P) = k(\theta)$ and let

$$\nu_3 : \mathbf{P}_k^1 \longrightarrow \mathbf{P}_k^3, \quad [s : t] \longmapsto [s^3 : s^2 t : s t^2 : t^3].$$

For $d = 4$, choose homogeneous coordinates $P = [a_0 : a_1 : a_2 : a_3]$ with $a_i \in E$. Noncoplanarity means that the a_i form a k -basis of E . Comparing this basis with $1, \theta, \theta^2, \theta^3$ gives a matrix in $\mathrm{GL}_4(k)$ taking $[1 : \theta : \theta^2 : \theta^3]$ to P . Its image of $\nu_3(\mathbf{P}^1)$ is the required curve.

For $d = 5$, let $\theta_1, \dots, \theta_5$ be the conjugates of θ in a Galois splitting field. The five points $[1 : \theta_i : \theta_i^2 : \theta_i^3]$ form a projective frame, by the Vandermonde determinant. The corresponding conjugates of P also form a projective frame. There is a unique projectivity matching these two labelled frames. Both labellings have the same Galois action, so uniqueness makes the projectivity Galois invariant; it therefore descends to k . Again its image of $\nu_3(\mathbf{P}^1)$ is a split twisted cubic Γ containing P .

If $\Gamma \subset S$, the curve supplies a k -point. Otherwise Bézout's theorem gives

$$\deg(S \cap \Gamma) = 9, \quad [S \cap \Gamma] - [P] \text{ effective of degree } 9 - d. \quad (2.1)$$

Under the standing assumption $S(k) = \emptyset$, this effective cycle is a single closed point, as above. $\square$

Remark 2.4. The effectivity argument in Voisin's proof of Proposition 4.3(b) [11] gives another proof. For $H = \mathcal{O}_S(2)$, one has $h^0(S, H) = 10$ and $c_1(H)^2 = 4h_3$. Add to a degree- d point a disjoint reduced line section. Voisin's Lemma 3.6, applied to the resulting subscheme of length $d + 3 \leq 8$, makes

$$4h_3 - ([P] + h_3) = 3h_3 - [P]$$

rationally equivalent to an effective zero-cycle of degree $9 - d$. Her proof explicitly uses $d = 5$; the same argument works for $d = 4$. Proposition 2.3 proves the existence equivalence by the direct intersection (2.1). It does not identify the resulting residual cycle with this particular Chow class.

3 Pfaffian cubics have rational points

Let $V = k^6$ and put $\mathcal{T} = (\wedge^2 V^*)^3 \simeq \mathbf{A}_k^{45}$. For an alternating form A , a subspace $W \subset V$ is isotropic if $A|_{W \times W} = 0$.

Lemma 3.1 (The degree-two isotropic incidence). *There is a nonempty open subset $\mathcal{T}^\circ \subset \mathcal{T}$, defined over k , such that the scheme of common isotropic three-spaces for a triple in $\mathcal{T}^\circ$ is finite étale of degree two.*

This is the ordered-triple form of Han's incidence calculation [7, Proposition 2.4]. The following proof also gives a fixed k -rational triple in $\mathcal{T}^\circ$.

Proof. Consider

$$\mathcal{I} = \{(W, A_0, A_1, A_2) : A_i|_{W \times W} = 0 \ (0 \leq i \leq 2)\} \subset \mathrm{Gr}(3, V) \times \mathcal{T}.$$

The restriction map $\wedge^2 V^* \rightarrow \wedge^2 W^*$ is surjective. Thus $\mathcal{I}$ is the total space of a rank-36 vector bundle over $\mathrm{Gr}(3, 6)$, and is smooth of dimension 45. The projection $\pi : \mathcal{I} \rightarrow \mathcal{T}$ is proper.

Write $V = V_1 \oplus V_2 \oplus V_3$, with $\dim V_i = 2$, and set

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Consider the triple

$$B_0 = \mathrm{diag}(J, J, J), \quad B_1 = \mathrm{diag}(J, 2J, 3J), \quad B_2 = \begin{pmatrix} 0 & I_2 & I_2 \\ -I_2 & 0 & D \\ -I_2 & -D & 0 \end{pmatrix}. \quad (3.1)$$

A B_0 -isotropic three-space is Lagrangian. If it is also B_1 -isotropic, it is invariant under $B_0^{-1}B_1 = \mathrm{diag}(I_2, 2I_2, 3I_2)$, hence splits into its three eigenspaces. Isotropy bounds the dimension of each summand by one, so $W = \ell_1 \oplus \ell_2 \oplus \ell_3$ with $\ell_i \subset V_i$ a line. This description holds scheme-theoretically: the spectral projectors split invariant subbundles after every base change. The common isotropic locus for B_0, B_1 is therefore $(\mathbf{P}^1)^3$.

Writing $\ell_i = [a_i : b_i]$, the remaining equations are

$$a_1a_2 + b_1b_2 = 0, \quad a_1a_3 + b_1b_3 = 0, \quad a_2a_3 + 2b_2b_3 = 0.$$

No a_i vanishes at a geometric solution: if $a_1 = 0$ then $b_2 = b_3 = 0$, contradicting the third equation; the other cases are similar. Set $u_i = b_i/a_i$. The fiber is

$$u_2 = u_3 = -u_1^{-1}, \quad u_1^2 + 2 = 0, \quad (3.2)$$

and hence is finite and geometrically reduced of degree two.

At these two geometric points the differential of π is an isomorphism, since source and target are smooth of the same dimension and the fiber tangent spaces vanish. Thus π is *étale* there. Properness allows us to remove the image of its non-*étale* locus and shrink around (B_0, B_1, B_2) ; the resulting morphism is proper and quasi-finite, hence finite *étale*, of degree two. $\square$

Proof of Theorem 1.2. Write

$$f(x) = \text{Pf} \left(\sum_{i=0}^3 x_i A_i \right),$$

where the A_i are alternating matrices over k . First assume $(A_0, A_1, A_2) \in \mathcal{T}^\circ(k)$. Lemma 3.1 gives an extension L/k of degree at most two and a three-space $W \subset L^6$ isotropic for A_0, A_1, A_2 .

The alternating form $A_3|_W$ has a nonzero radical vector $v \in W$, since $\dim W = 3$. Consequently all four covectors

$$A_0 v, A_1 v, A_2 v, A_3 v$$

belong to the three-dimensional subspace $\text{Ann}(W) \subset (L^6)^*$. Choose a nontrivial relation $\sum q_i A_i v = 0$ over L . The alternating matrix $\sum q_i A_i$ is singular, so

$$f(q_0, q_1, q_2, q_3)^2 = \det \left(\sum_{i=0}^3 q_i A_i \right) = 0.$$

This gives an L -point of $f = 0$. Its image has degree at most two, and Lemma 2.1 gives a k -point.

For an arbitrary triple, use the fixed matrices (3.1) and put

$$A_i(t) = (1-t)A_i + tB_i \quad (0 \leq i \leq 2), \quad A_3(t) = A_3, \quad f_t(x) = \text{Pf} \left(\sum_{i=0}^3 x_i A_i(t) \right).$$

The generic triple belongs to $\mathcal{T}^\circ$ because its value at $t = 1$ does; moreover $f_t \neq 0$ over $k(t)$ since $f_0 = f \neq 0$. The preceding argument over $k(t)$ gives a $k(t)$ -point of $f_t = 0$. Scale its homogeneous coordinates into the discrete valuation ring $k[t]_{(t)}$, with at least one a unit. Reduction modulo t then gives a k -point of $f = 0$. $\square$

The specialization applies to the final rational point; the auxiliary isotropic three-space need not specialize. Smoothness of the Pfaffian cubic is not used in this theorem.

4 Five points and the characteristic-zero theorem

Lemma 4.1 (Buchsbaum–Eisenbud, Beauville, Tauturri). *Let $Z \subset \mathbf{P}_k^3$ be a reduced subscheme of degree five whose geometric points are in general linear position. Every cubic form vanishing on Z has a linear 6×6 Pfaffian presentation over k .*

Proof. Put $R = k[x_0, x_1, x_2, x_3]$. The ideal of five points in general linear position is arithmetically Gorenstein, with resolution

$$0 \longrightarrow R(-5) \longrightarrow R(-3)^5 \longrightarrow R(-2)^5 \longrightarrow R \longrightarrow R/I_Z \longrightarrow 0. \quad (4.1)$$

For completeness, this standard five-point calculation can be checked after extending to $\bar{k}$ and moving the points to $e_0, e_1, e_2, e_3, [1 : 1 : 1 : 1]$. Their ideal is generated by the five independent differences among the six products $x_i x_j$ with $i < j$. In the quotient all mixed monomials of a fixed degree agree, and the Hilbert function is $1, 4, 5, 5, \dots$. Reducing by the nonzerodivisor $\ell = x_0 + x_1 + x_2 + x_3$, and writing q for the common mixed product, gives

$$x_i^2 = -3q, \quad x_i x_j = q \ (i \neq j), \quad x_i q = 0.$$

On the basis x_0, x_1, x_2 of degree one, multiplication into degree two is represented by

$$\begin{pmatrix} -3 & 1 & 1 \\ 1 & -3 & 1 \\ 1 & 1 & -3 \end{pmatrix}, \quad \det = -16.$$

Thus the socle is one-dimensional in degree two. The ring is Gorenstein, and its five quadratic generators and self-duality give (4.1). Gorensteinness and the graded Betti numbers descend by faithful flatness.

Apply the Buchsbaum–Eisenbud structure theorem [3] over k . The middle map of (4.1) is represented by a 5×5 alternating matrix T of linear forms, and I_Z is generated by

$$q_i = (-1)^{i+1} \operatorname{Pf}(T_i) \quad (1 \leq i \leq 5).$$

If f is a cubic vanishing on Z , write $f = \sum_i \ell_i q_i$ with linear forms ℓ_i over k . Expanding along the last column gives

$$f = \operatorname{Pf} \begin{pmatrix} T & \ell \\ -\ell^\top & 0 \end{pmatrix}, \quad \ell = (\ell_1, \dots, \ell_5)^\top. \quad (4.2)$$

This is the construction in [2, Remark 7.3(b)] and [10, Section 2]. The alternating resolution is obtained over k itself; no rationality of the individual geometric points is required. $\square$

Theorem 4.2. *A smooth cubic surface over a field of characteristic zero with an effective zero-cycle of degree five has a rational point.*

Proof. If the cycle has a point of degree one or two in its support, apply Lemma 2.1. Otherwise it is a single closed point P of degree five. The coplanar case is Lemma 2.2(1). In the remaining case Lemma 2.2(2) gives general linear position, Lemma 4.1 supplies a Pfaffian presentation over k , and Theorem 1.2 gives a rational point. $\square$

Corollary 4.3. *Theorem 1.1 holds in characteristic zero.*

Proof. Voisin’s theorem [11, Theorem 1.5] gives a point of degree one or four. In the latter case Proposition 2.3 gives an effective zero-cycle of degree five, and Theorem 4.2 applies. $\square$

There is also a universal formulation. Let $U \subset \mathbf{P}_k^{19}$ be the smooth-cubic locus, let $\mathcal{S} \rightarrow U$ be the universal cubic surface, and put $Y = \operatorname{Sym}_U^4(\mathcal{S})$. Over $k(Y)$ the universal surface has its tautological degree-four cycle. Proposition 2.3 and Theorem 4.2 give a rational section of

$$\mathcal{S} \times_U Y \longrightarrow Y.$$

Thus the construction works over the field of the unordered quartet; no ordering of its geometric points is needed.

5 Specialization to arbitrary fields

We now allow arbitrary characteristic. The required lifting statement is the smooth-surface instance of Ma’s method [8, Section 2.2, Lemmas 2.4–2.5].

Lemma 5.1. *Let k have characteristic $p > 0$, let $S \subset \mathbf{P}_k^3$ be a smooth cubic surface, and let $P \in S$ be a closed point of degree d . There exist a complete discrete valuation ring R of characteristic zero with residue field k , a smooth cubic surface $\mathcal{S}/R$ with special fiber S , and a finite flat subscheme $\mathcal{Z} \subset \mathcal{S}$ of degree d with special fiber P .*

Proof. Choose a Cohen ring R with uniformizer p and residue field exactly k [9, Tag 0328]. Lift the coefficients of a defining cubic of S to obtain $\mathcal{S} \subset \mathbf{P}_R^3$. This hypersurface is flat over R . It is smooth along its special fiber, and its nonsmooth locus is closed and proper over R . If this locus were nonempty, its image would contain the closed point of $\mathrm{Spec} R$. Thus $\mathcal{S}/R$ is smooth.

The closed immersion $P \hookrightarrow S$ is a local complete intersection: the maximal ideal of the regular local ring $\mathcal{O}_{S,P}$ is generated by a regular sequence. The locus of finite local complete intersections in $\mathrm{Hilb}^d(\mathcal{S}/R)$ is smooth over R [9, Tag 06D9]. Since R is complete, hence henselian, its k -point $[P]$ lifts to an R -point. The universal subscheme gives $\mathcal{Z}$. $\square$

One may instead use Fogarty's relative smoothness theorem for Hilbert schemes of points on a smooth family of surfaces over a discrete valuation ring [6, Theorem 2.9]. The local complete intersection argument isolates precisely what is needed here. In particular, $\kappa(P)/k$ need not be separable.

Proof of Theorem 1.1. By Corollary 4.3, only positive characteristic remains. Choose a closed point P of degree d prime to three from the support of a zero-cycle of degree one. Apply Lemma 5.1, and put $K = \mathrm{Frac}(R)$. The finite scheme $\mathcal{Z}_K$ has degree d , so its fundamental cycle, together with the degree-three hyperplane intersection class, gives a zero-cycle of degree one on $\mathcal{S}_K$.

Corollary 4.3 gives a K -point of $\mathcal{S}_K$. Properness extends it to an R -point of $\mathcal{S}$, whose reduction is a k -point of S . $\square$

The argument retains the original residue field and includes characteristics two and three. Only the final rational point is specialized; the Pfaffian presentation and the isotropic incidence are used in characteristic zero.

Proposition 5.2. *Independently of Theorem 1.2, Voisin's characteristic-zero degree reduction implies that a smooth cubic surface over an arbitrary field with a zero-cycle of degree one has a closed point of degree one or four.*

Proof. Lift a point of degree prime to three by Lemma 5.1 and apply Voisin's theorem to the generic fiber. A rational point specializes by properness. Otherwise a degree-four point gives a K -point of the projective scheme $\mathrm{Hilb}^4(\mathcal{S}/R)$. Its extension over R specializes to a finite subscheme of degree four on S .

If its cycle has a point of degree one, we are done. A degree-two point also gives a rational point in arbitrary characteristic: its scheme-theoretic linear span is a k -line, and the cubic intersection either contains that line or leaves a residual cycle of degree one. This includes purely inseparable quadratic points. If neither degree occurs, the effective degree-four cycle is a single closed point of degree four. $\square$

Remark 5.3. The same lifting and properness argument extends the existence equivalence of Proposition 2.3 to arbitrary fields. A finite subscheme of degree four or five on the generic fiber specializes through the corresponding projective Hilbert scheme. This transfers existence of effective cycles; it does not assert a positive-characteristic version of Voisin's Chow-group formulas.

References

- [1] Francesca Balestrieri, *Degrees of closed points on hypersurfaces*, Mathematische Nachrichten **297** (2024), no. 5, 1831–1837. [doi:10.1002/mana.202300160](https://doi.org/10.1002/mana.202300160).
- [2] Arnaud Beauville, *Determinantal hypersurfaces*, Michigan Mathematical Journal **48** (2000), 39–64. [doi:10.1307/mmj/1030132707](https://doi.org/10.1307/mmj/1030132707).
- [3] David A. Buchsbaum and David Eisenbud, *Algebra structures for finite free resolutions, and some structure theorems for ideals of codimension 3*, American Journal of Mathematics **99** (1977), no. 3, 447–485. [doi:10.2307/2373926](https://doi.org/10.2307/2373926).
- [4] Jean-Louis Colliot-Thélène, *Zéro-cycles sur les surfaces de del Pezzo (Variations sur un thème de Daniel Coray)*, L’Enseignement Mathématique **66** (2020), 447–487. [arXiv:2005.06876](https://arxiv.org/abs/2005.06876).
- [5] Daniel F. Coray, *Algebraic points on cubic hypersurfaces*, Acta Arithmetica **30** (1976), no. 3, 267–296. [doi:10.4064/aa-30-3-267-296](https://doi.org/10.4064/aa-30-3-267-296).
- [6] John Fogarty, *Algebraic families on an algebraic surface*, American Journal of Mathematics **90** (1968), no. 2, 511–521. [doi:10.2307/2373541](https://doi.org/10.2307/2373541).
- [7] Frédéric Han, *Pfaffian bundles on cubic surfaces and configurations of planes*, Mathematische Zeitschrift **278** (2014), nos. 1–2, 363–383. [doi:10.1007/s00209-014-1319-1](https://doi.org/10.1007/s00209-014-1319-1).
- [8] Qixiao Ma, *Closed points on cubic hypersurfaces*, Michigan Mathematical Journal **70** (2021), no. 4, 857–868. [doi:10.1307/mmj/1600329611](https://doi.org/10.1307/mmj/1600329611). [arXiv:1908.03139](https://arxiv.org/abs/1908.03139).
- [9] The Stacks Project Authors, *The Stacks Project*, [Tag 0328](#) (Cohen rings) and [Tag 06D9](#) (the finite local complete intersection Hilbert locus), accessed September 6, 2026.
- [10] Fabio Tanturri, *Pfaffian representations of cubic surfaces*, Geometriae Dedicata **168** (2014), 69–86. [doi:10.1007/s10711-012-9818-x](https://doi.org/10.1007/s10711-012-9818-x).
- [11] Claire Voisin, *Rank 2 vector bundles and degrees of points of del Pezzo surfaces*, Journal für die reine und angewandte Mathematik **836** (2026), 303–327. [doi:10.1515/crelle-2026-0025](https://doi.org/10.1515/crelle-2026-0025). [Author’s final version](#).